Reproduction report: *Persuasion in Parallel* (Coppock 2022)

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Archived at Zenodo: doi:10.5281/zenodo.24168318. This DOI always resolves to the latest release of the maintained version.

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1 What this repository is

This repository re-runs the empirical content of *Persuasion in Parallel: How Information Changes Minds about Politics* (Alexander Coppock, University of Chicago Press, 2022) against a current R environment, rewrites the analysis in modern idiom, and checks every number the maintained code produces against the number the book prints.

Item	Link
Book	https://doi.org/10.7208/chicago/9780226821832.001.0001
Publisher page	https://press.uchicago.edu/ucp/books/book/chicago/P/bo181475008.html
Replication archive	https://doi.org/10.7910/DVN/I9GSKI

```
coppock_2022/
  download_original.R          fetches and verifies the deposited archive
  run_all.R                    the entry point: sources everything, in order
  original/                    the deposited zip (not redistributed; gitignored)
  original_extracted/          the unpacked deposit (rebuilt on every run; gitignored)
  original_manifest.csv        the deposited file, its size and its published checksums
  original_members_manifest.csv  every member of the zip, by size and checksum
  maintained/                  the rewrite; one script per figure or table
  maintained/output/           everything the rewrite writes
  maintained/in_text_claims.R  the second instrument
  ground_truth/                the extraction, the build, and the comparison
  report/                      the rendered PDF of this document
```

To reproduce: clone the repository, open `coppock_2022.Rproj`, and run

```
source("run_all.R")
```

`run_all.R` fetches the archive from Dataverse, verifies it, runs all 28 analysis scripts, builds the ground truth, prints the in-text claims, and re-verifies the deposit. It took 9.4 minutes on the machine that produced the committed output; that figure describes one machine on one day and nothing else.

2 The book

Persuasion in Parallel argues that persuasive information moves the opinions of opposing groups in the same direction by similar amounts, and that the backlash effects predicted by motivated reasoning theory are rare. The evidence is 23 survey experiments, some original, some replications and reanalyses of published studies, reported across seven chapters and an appendix. Twelve of the experiments carry a panel component that supports the persistence analysis of chapter 6.

The book contains 67 numbered floats: 37 figures, 11 tables in the main text, and 19 in the appendix.

2.1 A limitation of this report, stated first

The only copy of the book available to this repository is the uncorrected proof. Every page carries the running foot *Uncorrected Proofs for Review Only*, dated 26 May 2022, and two running heads still read "Appendix 000" and "Notes to Pages 000-000" where the final page numbers had not been resolved. Published values in the ground truth are therefore transcribed from that proof rather than from the printed book.

The discrepancies the pipeline found are properties of the analysis rather than of the typesetting, so they do not depend on which copy is read, and `errata.qmd` states them with their corrected values computed at render time. Every one was checked against the printed edition, so none is an artifact of the proof or of the way a PDF's text layer is read. The seven reference and citation entries added later (errata 6 to 12) were not: they were read off the proof alone.

No errata page for the book exists on the publisher's site or on the author's own page.

3 The deposited archive

The deposit is a single 6 MB zip at https://doi.org/10.7910/DVN/I9GSKI, holding 30 R scripts, 21 pre-cleaned `.rds` data files and a README. Because the deposit is a container, `original/` holds one file and an emptiness check on it would be vacuous, so the repository carries two manifests: `original_manifest.csv` for the zip and `original_members_manifest.csv` for its 208 members. `download_original.R` verifies the zip against its published MD5 and its byte size, refuses to continue if `original/` holds anything the manifest does not list, audits `original_extracted/` before rebuilding it, rebuilds it from the zip, and audits it again. `run_all.R` sources that file first and again last, so a script that wrote into the deposit mid-run is caught at the end of the same run rather than months later.

The deposited checksum is 46790bddcaac7d2be07dab8e29358b1c, and it is the same value Dataverse publishes. Post-run checksums are not quoted anywhere in this report: a PDF records the time it was written, so a hash taken after a run can never recur.

3.1 Does the deposited code run?

Table 1: Deposited scripts by outcome, as shipped and stripped to data plus code

pass	clean	error
as_shipped	5	23
stripped	5	23

No. 23 of the 28 deposited analysis scripts fail, and 21 of them fail on one line for one reason: `vayr::sunflower()` replaced its `width` and `height` arguments with `density` and `aspect_ratio`, so every call in the deposit raises `unused arguments`. Since almost every figure in the book is a sunflower plot, that single API change takes down most of the archive. The remaining failure is `figure_3.1.R`, which dies inside `ggbrace::geom_brace()`.

The stripped-copy test is vacuous for this deposit and is recorded as such rather than reported as a pass: the archive ships 21 pre-cleaned `.rds` files and no derived object produced by its own code, so the stripped and as-shipped copies hold the same inputs and return the same per-script outcome.

The deposit writes nothing. There is no `ggsave()`, no `pdf()`, no `write_csv()` and no `png()` anywhere in its 30 scripts; the figures are drawn to the active device and the tables are printed to the console with `print.xtable()`. Running it therefore overwrites zero deposited files, and that measured zero has an explanation rather than being left to speak for itself: most scripts fail before reaching anything, and the ones that succeed have nothing to write. One stray file appears, an `Rplots.pdf` that `Rscript` leaves behind.

3.2 Why `value_script` is empty

The ground truth's archive column is empty on every row, and that is a finding rather than an omission. Twenty-three scripts never reach a value. Of the five that run, three print nothing at all, and the two that do print (the chapter 6 table scripts) call `rsample::bootstraps()` with no seed, so what they print changes from run to run and could not honestly be committed as a property of the deposit.

4 Chapter 6 is unseeded

Neither of the deposit's two chapter 6 scripts sets a seed. Every number in tables 6.2, 6.3 and 6.4, in figures 6.1 and 6.2, and in the persistence sentences of chapter 6 is one draw of an unseeded bootstrap. The bootstrap reaches the point estimates as well as the standard errors, which is easy to miss: `metafor::rma()` weights each study by the inverse of its squared standard error, and those standard errors are the bootstrapped ones.

`ground_truth/measure_seed_dispersion.R` re-runs both scripts across 20 seeds and records how far each published quantity could have landed. Rows that depend on it carry `match_rewrite = NA` with the verdict in `holds`, compared against the range rather than against a single number.

Table 2: Range of each chapter 6 quantity across seeds

Float	Quantity	Minimum	Median	Maximum
table_6.2	Overall	0.340	0.340	0.350
table_6.2	Prediction: stronger persistence	0.500	0.500	0.510
table_6.2	Prediction: weaker persistence	0.210	0.210	0.210
table_6.3	Democrat, 10 days	0.470	0.480	0.500
table_6.3	Democrat, 30 days	0.460	0.470	0.490
table_6.3	Overall, 10 days	0.470	0.480	0.480
table_6.3	Overall, 30 days	0.450	0.450	0.460
table_6.3	Republican, 10 days	0.510	0.520	0.530
table_6.3	Republican, 30 days	0.560	0.560	0.580

5 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: every claim the comparison covers, with the location in the book that states it, its type, the string the page prints, and the number of decimals it prints at. It carries 111 rows, of which 80 are prose claims that must have a block in `maintained/in_text_claims.R` and 31 are float-level rows reading cells-reproduced-of-cells.

Two instruments reach the same claimed numbers by separate paths from the same pipeline outputs:

- `ground_truth/build_ground_truth.R` joins each published value to what `maintained/output/` contains and writes `coppock_2022_ground_truth.csv`.
- `maintained/in_text_claims.R` carries the book's sentence verbatim beside code that recomputes the number, and prints `CLAIM <id> = <value> || [verdict] <label>`, scored by `excheckr` against the extraction and the errata spine.

Where a second derivation exists the claims file takes it: the chapter 1 treatment effects are recovered there by differencing the two condition means the figure plots, and the chapter 5 correlations by recomputing them from the plotted pairs, rather than by reading back the same file the build reads.

The gate in `build_ground_truth.R` runs the claims file non-interactively under `capture.output()`, in its own environment, and asserts that the number of printed claim ids equals the number of required blocks, that every required id is printed, that no printed id is undeclared, and that the value each block printed equals the value the build rendered. A separate check re-renders each published string at the digit count the extraction records and requires it to equal the string stored, so a shared `digits` column cannot make the two instruments agree about a precision that is wrong.

The gate was tested by breaking it rather than by reading it. Deleting a block fails the count; changing a value in a block fails the cross-instrument comparison; changing a `digits` entry fails the precision check.

5.1 What the extraction does not cover

The extraction covers the numeric claims of chapters 1 through 6 and every float in the book. It deliberately excludes three classes, each for a reason that was checked rather than assumed:

- The appendix reproduces the full text of five newspaper op-eds and the wording of every treatment and outcome used in the 23 experiments. Those pages are dense with numbers ("$9 billion in debt", "14.5% flat tax", "8 of the 10 pairs"), and every one is transcribed from a source outside the book. They cannot drift and nothing in the pipeline could check them.
- Chapters 3, 4 and 7 contain worked illustrations: a hypothetical 3-point and 5-point effect, a power calculation returning 3,000 subjects, a simulated subject with ten considerations, and a run of Bayes' rule arithmetic. These are definitional, and each is verified where it is used rather than against a pipeline that never computes them.
- Tables 2.1 and the upper half of table 2.3 reproduce Lord, Ross, and Lepper's own tables 3 and 1 from 1979. They are transcriptions, checked once against that article.

6 Ground truth

Table 3: Ground truth verdicts

Verdict	Rows
Does not hold	8
Does not match	6

Holds (no printed value to match)	12
Matches at the book's precision	80
No verdict (approximate, or nothing computed)	5

Table 4: Every row that does not agree

Claim
Robust standard error for the flat tax op-ed effect for Republicans among policy professionals
Slope of support for gay marriage per year among Democrats
Standard error of the two-sided effect on support among opponents
Average persistence ratio in the weaker persistence group in percent
Persistence of the Hiscox expert treatment on Lucid in percent
Persistence ratios above 60 percent for the pro minimum wage videos the flat tax op-ed and the pro capital p
Overall op-ed persistence ten days after treatment in percent
Overall op-ed persistence thirty days after treatment in percent
Op-ed persistence ten days after treatment for Republicans in percent
Op-ed persistence ten days after treatment for Democrats in percent
Cells of appendix table A.1 reproduced
Cells of appendix table A.8 reproduced
Cells printed under After pro-capital punishment study in table 2.2 that are pro-study means
Cells of the Guess and Coppock half of table 2.3 reproduced

6.1 Float coverage

Across the book's 67 floats the pipeline reproduces 1,054 of 1,390 published values. The fraction is stated for every float that prints one, because an aggregate hides a float covered by a single row:

Table 5: Published values reproduced, by float

Float	Published values	Reproduced	Fraction
Table 2.1	12	12	100%
Table 2.2	12	12	100%
Table 2.3	24	23	96%
Table 4.2	23	0	0%
Table 6.1	12	0	0%
Table 6.2	9	9	100%
Table 6.3	12	12	100%
Table 6.4	114	114	100%

Table A.1	108	0	0%
Table A.10	144	144	100%
Table A.12	132	132	100%
Table A.14	44	44	100%
Table A.16	44	44	100%
Table A.18	32	32	100%
Table A.19	44	44	100%
Table A.2	24	24	100%
Table A.4	48	48	100%
Table A.5	120	120	100%
Table A.6	72	72	100%
Table A.7	48	48	100%
Table A.8	192	0	0%
Table A.9	120	120	100%

4 published floats have zero coverage, and each has a reason that was tested:

Table 6: Published floats with no coverage

Float	Published values	Reason
Table 4.2	23	Sample sizes for the 23 experiments. The deposit computes none of them and
Table 6.1	12	Panel sample sizes for the 12 persistence experiments, on the same footing as
Table A.1	108	The deposit contains no code for the Coppock, Ekins, and Kirby treatment ef
Table A.8	192	The deposit contains no code for the Hiscox treatment effect estimates.

The reason given for tables A.1 and A.8 is a measurement rather than an impression. The deposit contains thirteen `print.xtable()` calls; eleven build appendix tables and two build the chapter 6 tables, and none of the thirteen builds A.1 or A.8. A reconstruction of A.8 from the deposited data reproduces its valence contrasts and not all of its expert contrasts, which is why the specification is treated as unrecoverable rather than guessed at.

The 37 figures print no estimates on their faces, so they contribute nothing to the counts above. Every one is a scatter of individual responses with condition means and intervals drawn over it, so what a figure asserts is the set of estimates it plots, and each figure script writes those estimates to a CSV: 1,435 estimates across the 30 figures the rewrite draws. Seven figures have no such file: figure 1.1 and figures 4.1, 4.2 and 7.1 are schematics and simulations with no data behind them, figure 2.1 reproduces two graphics used as experimental stimuli, figure 3.4 is a diagram, and figure 5.14 reproduces the two income-inequality plots that are themselves the Trump and White treatment.

6.2 The appendix

`ground_truth/published_appendix_values.csv` holds every cell of the book's 14 appendix estimate tables, 1172 published numbers in 293 rows, parsed positionally from the page. Positional parsing is not a preference here: read in reading order, `pdftotext` loses the numeric columns of table A.2 entirely and returns a table that looks complete and is not. Three artifacts of this PDF each drop rows if unhandled: a negative number is typeset as a minus sign followed by a soft hyphen and sometimes a space, the minus sign is U+2212 rather than a hyphen, and the small-capital T of "Table" is dropped from table A.10's caption.

7 Errata

Nine sentences, labels or cells in the book do not survive the reanalysis, in five entries, and seven more entries correct the reference list and the citations, for twelve in all, together with 3 notes that correct nothing (entries 13 to 15, below). They are set out in `errata.qmd`, rendered to `coppock_2022_errata.pdf` at the repository root, with every corrected value computed at render time: from `maintained/output/` where the reanalysis settles it, and from the book's own transcribed tables where the estimator behind it is unseeded. Summarized here:

1. Table 2.2's two study rows carry each other's labels, in both panels. The cells printed under "After pro-capital punishment study" are the means among subjects who read the anti-capital punishment study, and the reverse. Three artifacts identify which row is which and all three agree: the content variable in the deposited data, figure 2.2 drawn from that same variable, and the sentence on page 26 reporting that self-assessed change is higher in the pro condition than in the null and higher in the null than in the con. The values are correct and are printed in the order the analysis produces them; the "Combined" rows are sums and are unaffected, which is why the reading the surrounding text draws from the table is unchanged.

2. Four sentences in chapter 6 report persistence figures that the chapter's own tables print differently. Page 114 gives the weaker persistence group as 20 per cent where table 6.2 prints 0.21; page 117 gives the op-ed effects as 46 per cent after ten days and 44 after thirty where table 6.3 prints 0.48 and 0.46; page 118 gives 57 per cent for Republicans and 43 for Democrats after ten days where table 6.3 prints 0.53 and 0.48. Every persistence figure in the chapter comes from a bootstrap the deposit runs without a seed, so no rerun settles what the sentences should say. The tables do, and they are also the side the estimator supports: every figure the tables print is inside the range its quantity takes across seeds, and every figure the prose gives is outside it.

Table 7: Chapter 6 prose against the chapter 6 tables

Sentence	In the text	In its own table	Across seeds
p. 114, weaker persistence group	20%	21%	21% to 21%

p. 117, op-eds after ten days	46%	48%	47% to 48%
p. 117, op-eds after thirty days	44%	46%	45% to 46%
p. 118, Republicans after ten days	57%	53%	51% to 53%
p. 118, Democrats after ten days	43%	48%	47% to 50%

The "In its own table" column is the correction the errata prints. It is not recomputed, which is the one thing the unseeded estimator forbids; it is the figure the book's own table already carries, and it comes from the transcription in `ground_truth/published_maintext_tables.csv` rather than from a run.

3. Page 114 attributes a persistence estimate to a sample that is not in the analysis. The sentence reads "the different replications of the Hiscox (2006) 'expert' treatment generated very different persistence estimates: 49 percent on MTurk but 0 percent on Lucid." No Lucid sample appears anywhere in the persistence analysis. Table 6.1 lists twelve panel experiments and the Hiscox rows of table 6.4 are Mechanical Turk and GfK. The book's own table 6.4 gives the zero to GfK, and the pipeline reproduces it there.

4. Table 2.3 contradicts itself. In the lower half, the mean rating of how convincing the two studies were is 3.4 for the pro-capital punishment study and 0.1 for the anti-capital punishment study among proponents, and the table prints both. It prints their difference as 3.0, where the two cells differ by 3.3. Every other cell of that half reproduces.

5. Page 101 disagrees with appendix table A.5. The sentence gives the standard error of the two-sided message's effect on capital punishment support among opponents as 0.11. Table A.5 gives 0.12 for the same quantity, and the pipeline reproduces the appendix.

6 to 12. The reference list and the citations. An audit checking every entry in the reference list against its published record found four defects there: Lord's middle initial (S. for G.), Nyhan et al.'s first page (936 for 939), and the titles of Blair, Coppock, and Moor (2020) and of Green, Wilke, and Cooper (2020), each printed as an earlier draft's. A citation cross-check found three more in the text: "Guess and Coppock (2018)" in four study tables against the list's 2020, "Little, Schnakenberg, and Turner (2020)" on page 127 against 2021, and "Azjen" on page 42. These seven were read off the proof only.

13. The Democratic slope in the gay marriage series is given as 2.1 points per year. Page 11 says "Democrats 2.1 points per year", and the reanalysis of figure 1.4 returns 2.02, which prints 2.0. The other three slopes that sentence names reproduce exactly. Entry 13 is a note that corrects nothing, for the reason the examination gave: a single slope off by a tenth in a sentence whose point is that the slopes are similar does not mislead a reader who quotes it, and no replacement can be set without guessing at the author's number.

14. A standard error for Republican policy professionals is given as 0.25. The flat tax effect among Republican policy professionals has a standard error of 0.24453 under HC2, which prints 0.24 against a published 0.25; under HC3 it is 0.24589 and prints 0.25, and HC3 reproduces all eight of that chapter's standard errors where HC2 reproduces seven. The

deposit contains no code for any of the eight, so the variance estimator behind the published figure cannot be established, and the rewrite keeps the house default rather than the one that happens to match. Entry 14 is a note that corrects nothing, because which estimator the author used is the open question.

15. "All had persistence ratios above 60 percent" names no sample. Page 114 says the pro-minimum wage videos, the flat tax op-ed and the pro-capital punishment studies "all had persistence ratios above 60 percent". Those three treatments span four rows of table 6.4, because the flat tax op-ed was fielded on two samples and the sentence names neither. Three of the four clear 60 per cent; the flat tax op-ed among policy professionals is 39 per cent in the book's own table 6.4 and 39 per cent here. The sentence holds on Mechanical Turk, where all three treatments were run, so what is wrong is not a number but the absence of one word, and the ground truth records it as unresolved. Entry 15 is a note that corrects nothing, because the missing word is the author's to supply.

Entries 13 to 15 are the three further discrepancies that were examined and are not errata. Each is carried in `errata.qmd` as a note, which quotes the passage as the proof prints it and sets no corrected side.

8 The maintained rewrite

`maintained/` holds 28 analysis scripts and a `helpers.R`, one script per figure, with an appendix table folded into the figure script that shares its model where the two are inseparable. Every script sources `helpers.R` first and loads no packages of its own. Paths go through `here::here()`. Output goes to `maintained/output/`, which is committed so a reader can diff a fresh run against it.

The substitutions the current environment forced:

Deposit	Rewrite	Why
`vayr::sunflower(x, y, width, height)`	`sunflower_compat()`	vayr 1.0.0 replaced the two semi-axis arguments with `density` and `aspect_ratio`; the helper inverts the new API's own formulas so the deposit's layout is recovered exactly
`do(tidy(...))`	`reframe(tidy(...))`	`do()` is superseded
`melt()` / `dcast()`	`pivot_longer()` / `pivot_wider()`	reshape2 is superseded
`geom_errorbar(width = 0)`	`geom_linerange()`	house style for interval bars

Deposit	Rewrite	Why
`xtable()` + `print.xtable()`	`write_csv()`	a table printed to the console cannot be diffed
`rmeta::meta.summaries()`	`metafor::rma()`	unmaintained since 2018, and its non-standard evaluation fails inside `reframe()`
`haven_labelled` outcome	`haven::zap_labels()`	`sunflower()` cannot do arithmetic on a labelled vector

Six defects in this repository's own code were found by the comparison and fixed:

- Its Brader, Valentino, and Suhay appendix table (A.7) filtered independents out of the data at load, so the pooled "Overall" rows were fitted on Democrats and Republicans alone. The deposit filters them from the printed rows and not from the sample. All 13 affected cells now reproduce.
- Its figure 5.16 script computed the six correlations chapter 5 quotes with six bare `with(...)` calls at the top level, which print nothing when a file is sourced and wrote nothing. The figure's headline quantities had no committed output at all.
- Its figure 2.3 script carried a comment saying table 2.3 reports only the original 1979 study and could not be produced. Half of that was false: the deposited data carry the replication's own quality and convincingness ratings, and computing them is what surfaced the contradiction in that table.
- `ground_truth/run_archive.R` described every clean script as having exited non-zero, because `tibble()` evaluates its arguments in order and a `status` column shadowed the exit code the next argument read.
- `ground_truth/build_ground_truth.R` normalized signed zero on the rewrite's side of the appendix comparison and not on the book's, so the two appendix cells the book prints as `-0.00` were counted as failures against a rewritten `0.00`.
- Its figure 5.3 script scattered the raw responses of two panels with an unseeded `runif()`, following the deposit, so the figure was different on every run while its estimates were not. The run-twice diff is what surfaced it, and it is the only file in `maintained/output/` that was ever not byte-reproducible.

9 Verification

9.1 The figures

Every rendered figure was laid beside the published page. The check is not redundant with the numeric one: a transposed axis or a swapped facet leaves every estimate correct and every

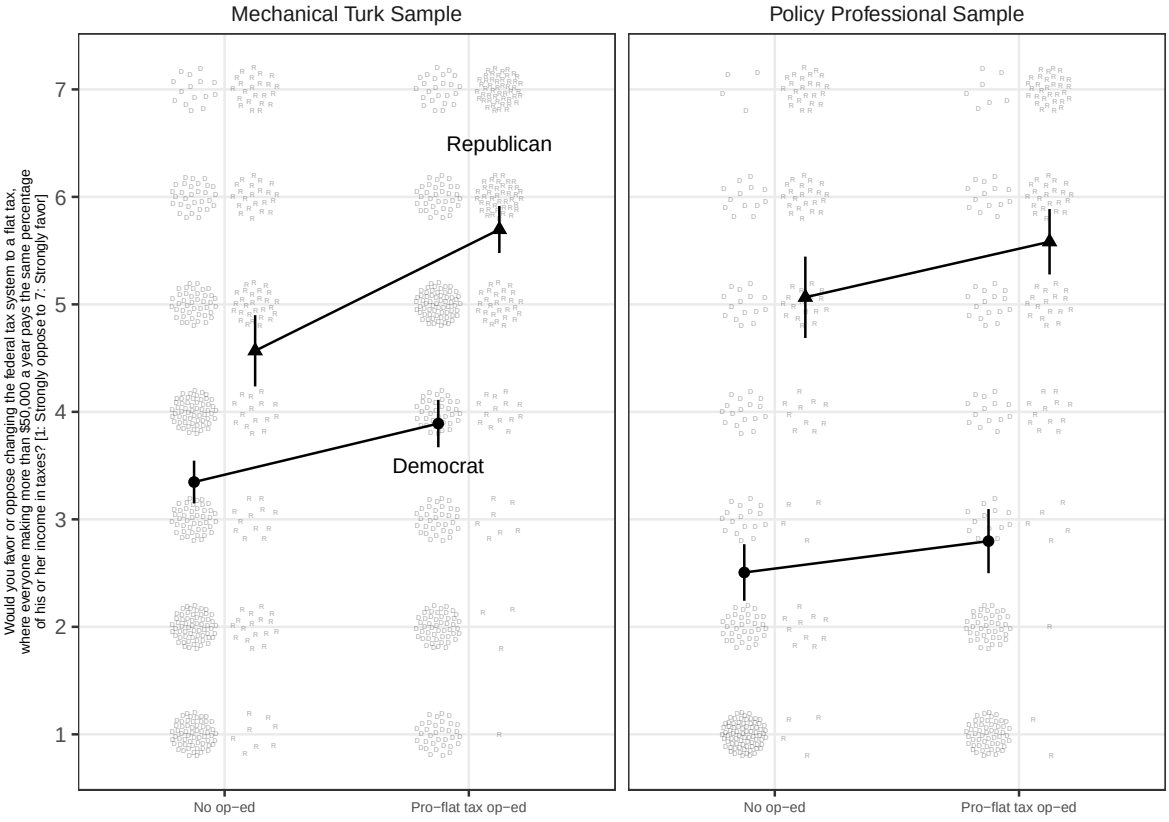


Figure 1: Figure 1.2, the flat tax op-ed experiment, as the rewrite draws it.

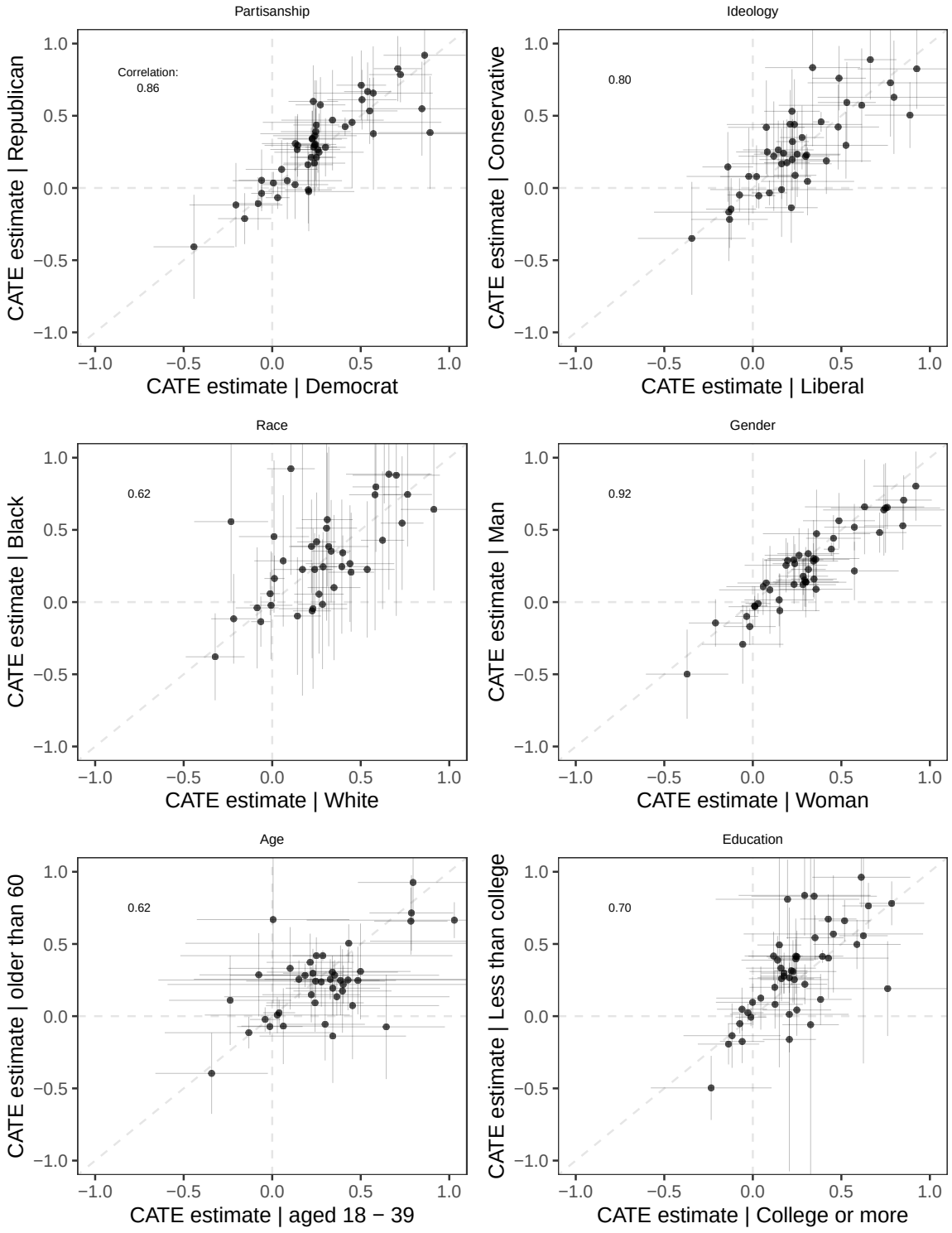


Figure 2: Figure 5.16, conditional average treatment effects across subgroups, as the rewrite draws it.

value comparison passing. All 30 reproduced figures match the published layout, including the facet order and the direction of every axis.

9.2 The pipeline

`run_all.R` was run to completion from clean sessions and the whole of `maintained/output/` diffed between runs. Every CSV and PNG comes back byte-identical, as does `ground_truth/coppock_2022_ground_truth.csv`. The figure PDFs differ, and only in their embedded `CreationDate` and `ModDate`, which a PDF records on every write.

The one thing this check found is worth naming, because it is exactly what the check is for: figure 5.3 came back different on the second run. Its estimates were identical and its CSV was byte-identical; what moved was the position of the faint background points in two of its four panels, scattered by an unseeded `runif()`. Nothing in the numeric comparison could have seen it.

10 R environment

Table 8: Environment the committed output was produced under

Component	Version
R	4.6.2
tidyverse	2.0.0
estimatr	2.0.1
metafor	5.2.1
vayr	1.1.0
ggplot2	4.0.3
rsample	1.3.2

11 Licence

CC0 1.0, matching the deposit.